

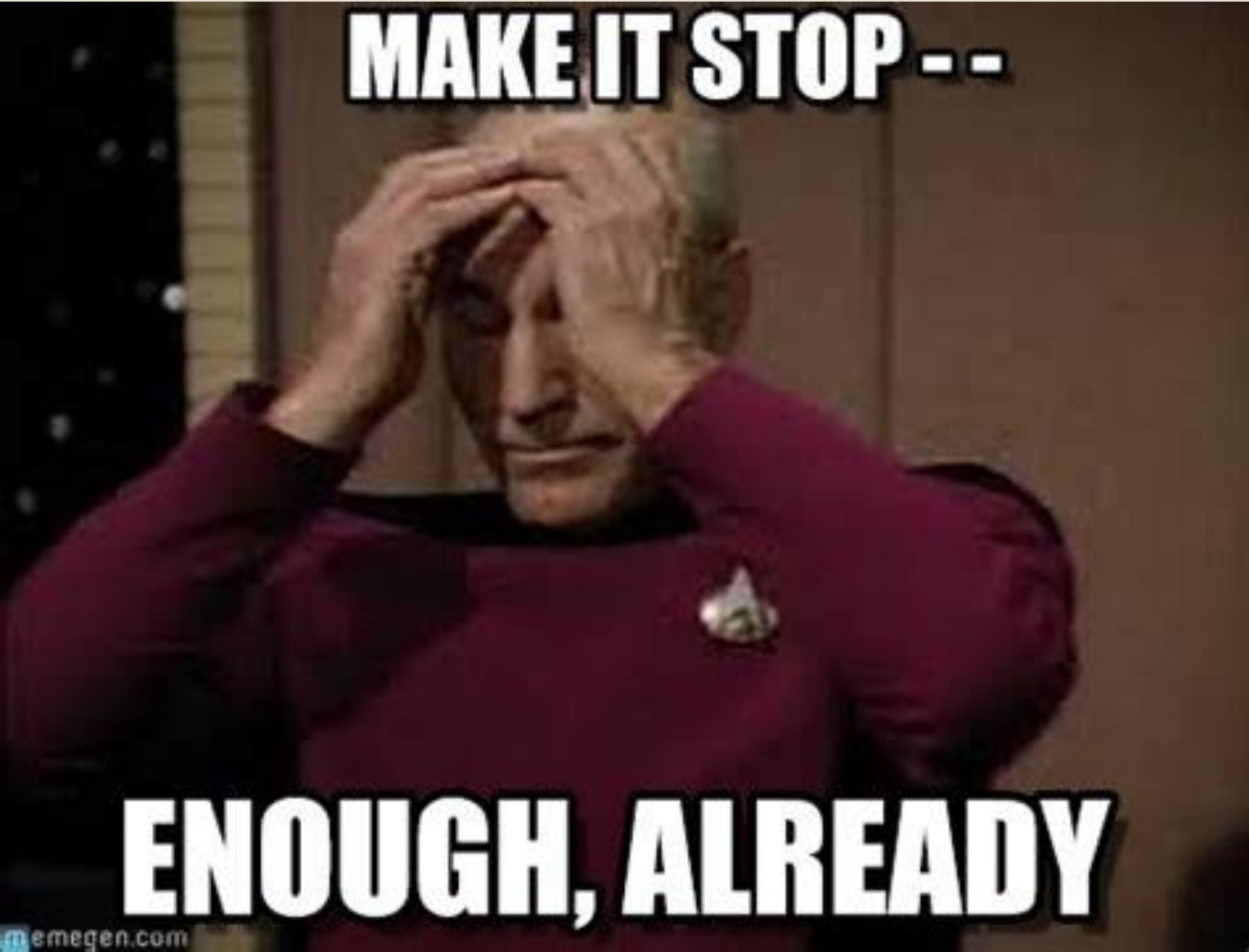
WHY BE NORMAL?



order_id	order_date	cust_id	cust_name	cust_state	item_id	item_desc	item_quant	item_price	value
1001	10/13	1004	Mo	OR	6531, 7814, 2305	Tape, Paper, Pens	4, 5, 12	2.00, 10.00, 6.00	8.00, 50.00, 72.00
1002	10/12	2005	Larry	CA	7814, 7301	Paper, Erasers	10, 5	10.00, 1.00	100.00, 5.00
1003	10/17	3006	Curly	WA	4234, 2305	Pencils, Pens	12, 9	6.00, 6.00	72.00, 54.00
1004	10/19	1004	Mo	OR	6531, 2305	Tape, Pens	5, 9	2.00, 6.00	10.00, 54.00

```
create table normalization_activity.cust_orders (  
  order_id int  
  , order_date date  
  , cust_id int  
  , cust_name text  
  , cust_state text  
  , item_id text  
  , item_desc text  
  , item_quant text  
  , item_price text  
  , value text  
  , primary key (order_id)  
) ;
```

Yes, we could make some of these arrays, but the performance and storage of arrays in a tables causes other issues.



select * from r	
order_id	order_date
1,001	2023-10-
1,002	2023-10-
1,003	2023-10-
1,004	2023-10-

em_price	value
.00,10.0 ,6.00	8.00,50.0 0,72.00
0.00,1.0	100.00,5. 00
.00,6.00	72.00,54. 00
.00,6.00	10.00,54. 00

order_id	order_date	cust_id	cust_name	cust_state	item_id	item_desc	item_quant	item_price	value
1001	10/13	1004	Mo	OR	6531, 7814, 2305	Tape, Paper, Pens	4, 5, 12	2.00, 10.00, 6.00	8.00, 50.00, 72.00
1002	10/12	2005	Larry	CA	7814, 7301	Paper, Erasers	10, 5	10.00, 1.00	100.00, 5.00
1003	10/17	3006	Curly	WA	4234, 2305	Pencils, Pens	12, 9	6.00, 6.00	72.00, 54.00
1004	10/19	1004	Mo	OR	6531, 2305	Tape, Pens	5, 9	2.00, 6.00	10.00, 54.00

Repeating customer data

- Customer id
- Customer name
- Customer state

order_id	order_date	cust_id	cust_name	cust_state	item_id	item_desc	item_quant	item_price	value
1001	10/13	1004	Mo	OR	6531, 7814, 2305	Tape, Paper, Pens	4, 5, 12	2.00, 10.00, 6.00	8.00, 50.00, 72.00
1002	10/12	2005	Larry	CA	7814, 7301	Paper, Erasers	10, 5	10.00, 1.00	100.00, 5.00
1003	10/17	3006	Curly	WA	4234, 2305	Pencils, Pens	12, 9	6.00, 6.00	72.00, 54.00
1004	10/19	1004	Mo	OR	6531, 2305	Tape, Pens	5, 9	2.00, 6.00	10.00, 54.00

Repeating order data

- Order id
- Order date
- Item id

order_id	order_date	cust_id	cust_name	cust_state	item_id	item_desc	item_quant	item_price	value
1001	10/13	1004	Mo	OR	6531, 7814, 2305	Tape, Paper, Pens	4, 5, 12	2.00, 10.00, 6.00	8.00, 50.00, 72.00
1002	10/12	2005	Larry	CA	7814, 7301	Paper, Erasers	10, 5	10.00, 1.00	100.00, 5.00
1003	10/17	3006	Curly	WA	4234, 2305	Pencils, Pens	12, 9	6.00, 6.00	72.00, 54.00
1004	10/19	1004	Mo	OR	6531, 2305	Tape, Pens	5, 9	2.00, 6.00	10.00, 54.00

Repeating item data

- Item description
- Item price

order_id	order_date	cust_id	cust_name	cust_state	item_id	item_desc	item_quant	item_price	value
1001	10/13	1004	Mo	OR	6531, 7814, 2305	Tape, Paper, Pens	4, 5, 12	2.00, 10.00, 6.00	8.00, 50.00, 72.00
1002	10/12	2005	Larry	CA	7814, 7301	Paper, Erasers	10, 5	10.00, 1.00	100.00, 5.00
1003	10/17	3006	Curly	WA	4234, 2305	Pencils, Pens	12, 9	6.00, 6.00	72.00, 54.00
1004	10/19	1004	Mo	OR	6531, 2305	Tape, Pens	5, 9	2.00, 6.00	10.00, 54.00

Redundant data

- value

order_id	order_date	cust_id	cust_name	cust_state	item_id	item_desc	item_quant	item_price	value
1001	10/13	1004	Mo	OR	6531, 7814, 2305	Tape, Paper, Pens	4, 5, 12	2.00, 10.00, 6.00	8.00, 50.00, 72.00
1002	10/12	2005	Larry	CA	7814, 7301	Paper, Erasers	10, 5	10.00, 1.00	100.00, 5.00
1003	10/17	3006	Curly	WA	4234, 2305	Pencils, Pens	12, 9	6.00, 6.00	72.00, 54.00
1004	10/19	1004	Mo	OR	6531, 2305	Tape, Pens	5, 9	2.00, 6.00	10.00, 54.00

order_id	order_date	cust_id	item_id	item_desc	item_quant	item_price	value
1001	10/13	1004	6531, 7814, 2305	Tape, Paper, Pens	4, 5, 12	2.00, 10.00, 6.00	8.00, 50.00, 72.00
1002	10/12	2005	7814, 7301	Paper, Erasers	10, 5	10.00, 1.00	100.00, 5.00
1003	10/17	3006	4234, 2305	Pencils, Pens	12, 9	6.00, 6.00	72.00, 54.00
1004	10/19	1004	6531, 2305	Tape, Pens	5, 9	2.00, 6.00	10.00, 54.00

customers		
cust_id	cust_name	cust_state
1004	Mo	OR
2005	Larry	CA
3006	Curly	WA
1004	Mo	OR

order_id	order_date	cust_id	item_id	item_desc	item_quant	item_price	value
1001	10/13	1004	6531, 7814, 2305	Tape, Paper, Pens	4, 5, 12	2.00, 10.00, 6.00	8.00, 50.00, 72.00
1002	10/12	2005	7814, 7301	Paper, Erasers	10, 5	10.00, 1.00	100.00, 5.00
1003	10/17	3006	4234, 2305	Pencils, Pens	12, 9	6.00, 6.00	72.00, 54.00
1004	10/19	1004	6531, 2305	Tape, Pens	5, 9	2.00, 6.00	10.00, 54.00

customers		
cust_id	cust_name	cust_state
1004	Mo	OR
2005	Larry	CA
3006	Curly	WA
1004	Mo	OR

order_id	item_id	item_desc	item_quant	item_price	value
1001	6531, 7814, 2305	Tape, Paper, Pens	4, 5, 12	2.00, 10.00, 6.00	8.00, 50.00, 72.00
1002	7814, 7301	Paper, Erasers	10, 5	10.00, 1.00	100.00, 5.00
1003	4234, 2305	Pencils, Pens	12, 9	6.00, 6.00	72.00, 54.00
1004	6531, 2305	Tape, Pens	5, 9	2.00, 6.00	10.00, 54.00

customers		
cust_id	cust_name	cust_state
1004	Mo	OR
2005	Larry	CA
3006	Curly	WA
1004	Mo	OR

orders		
order_id	order_date	cust_id
1001	10/13	1004
1002	10/12	2005
1003	10/17	3006
1004	10/19	1004

order_id	item_id	item_desc	item_quant	item_price	value
1001	6531, 7814, 2305	Tape, Paper, Pens	4, 5, 12	2.00, 10.00, 6.00	8.00, 50.00, 72.00
1002	7814, 7301	Paper, Erasers	10, 5	10.00, 1.00	100.00, 5.00
1003	4234, 2305	Pencils, Pens	12, 9	6.00, 6.00	72.00, 54.00
1004	6531, 2305	Tape, Pens	5, 9	2.00, 6.00	10.00, 54.00

customers		
cust_id	cust_name	cust_state
1004	Mo	OR
2005	Larry	CA
3006	Curly	WA
1004	Mo	OR

orders		
order_id	order_date	cust_id
1001	10/13	1004
1002	10/12	2005
1003	10/17	3006
1004	10/19	1004

order_id	item_id	item_quant	item_price	value
1001	6531,	4,	2.00,	8.00,
	7814,	5,	10.00,	50.00,
	2305	12	6.00	72.00
1002	7814,	10,	10.00,	100.00,
	7301	5	1.00	5.00
1003	4234,	12,	6.00,	72.00,
	2305	9	6.00	54.00
1004	6531,	5,	2.00,	10.00,
	2305	9	6.00	54.00

customers		
cust_id	cust_name	cust_state
1004	Mo	OR
2005	Larry	CA
3006	Curly	WA
1004	Mo	OR

orders		
order_id	order_date	cust_id
1001	10/13	1004
1002	10/12	2005
1003	10/17	3006
1004	10/19	1004

items		
item_id	item_desc	item_price
6531	Tape	2.00
7814	Paper	10.00
2305	Pens	6.00
7301	Erasers	1.00
4234	Pencils	6.00

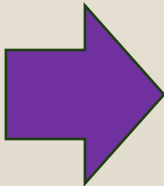
order_id	item_id	item_quant	item_price	value
1001	6531,	4,	2.00,	8.00,
	7814,	5,	10.00,	50.00,
	2305	12	6.00	72.00
1002	7814,	10,	10.00,	100.00,
	7301	5	1.00	5.00
1003	4234,	12,	6.00,	72.00,
	2305	9	6.00	54.00
1004	6531,	5,	2.00,	10.00,
	2305	9	6.00	54.00

customers		
cust_id	cust_name	cust_state
1004	Mo	OR
2005	Larry	CA
3006	Curly	WA
1004	Mo	OR

orders		
order_id	order_date	cust_id
1001	10/13	1004
1002	10/12	2005
1003	10/17	3006
1004	10/19	1004

items		
item_id	item_desc	item_price
6531	Tape	2.00
7814	Paper	10.00
2305	Pens	6.00
7301	Erasers	1.00
4234	Pencils	6.00

order_id	item_id	item_quant	item_price	value
1001	6531,	4,	2.00,	8.00,
	7814,	5,	10.00,	50.00,
	2305	12	6.00	72.00
1002	7814,	10,	10.00,	100.00,
	7301	5	1.00	5.00
1003	4234,	12,	6.00,	72.00,
	2305	9	6.00	54.00
1004	6531,	5,	2.00,	10.00,
	2305	9	6.00	54.00



order_items			
order_id	item_id	item_quant	price_paid
1001	6531	4	2.00
1001	7814	5	10.00
1001	2305	12	6.00
1002	7814	10	10.00
1002	7301	5	1.00
1003	4234	12	6.00
1003	2305	9	6.00
1004	6531	5	2.00
1004	2305	9	6.00

customers		
cust_id	cust_name	cust_state
1004	Mo	OR
2005	Larry	CA
3006	Curly	WA
1004	Mo	OR

orders		
order_id	order_date	cust_id
1001	10/13	1004
1002	10/12	2005
1003	10/17	3006
1004	10/19	1004

items		
item_id	item_desc	item_price
6531	Tape	2.00
7814	Paper	10.00
2305	Pens	6.00
7301	Erasers	1.00
4234	Pencils	6.00

customers		
cust_id	cust_name	cust_state
1004	Mo	OR
2005	Larry	CA
3006	Curly	WA
1004	Mo	OR

order_items			
order_id	item_id	item_quant	price_paid
1001	6531	4	2.00
1001	7814	5	10.00
1001	2305	12	6.00
1002	7814	10	10.00
1002	7301	5	1.00
1003	4234	12	6.00
1003	2305	9	6.00
1004	6531	5	2.00
1004	2305	9	6.00

orders		
order_id	order_date	cust_id
1001	10/13	1004
1002	10/12	2005
1003	10/17	3006
1004	10/19	1004

items		
item_id	item_desc	item_price
6531	Tape	2.00
7814	Paper	10.00
2305	Pens	6.00
7301	Erasers	1.00
4234	Pencils	6.00

customers		
cust_id	cust_name	state_id
1004	Mo	1
2005	Larry	2
3006	Curly	3
1004	Mo	1

states		
state_id	state_abbrev	state_name
1	OR	Oregon
2	CA	California
3	WA	Washington

orders		
order_id	order_date	cust_id
1001	10/13	1004
1002	10/12	2005
1003	10/17	3006
1004	10/19	1004

order_items			
order_id	item_id	item_quant	price_paid
1001	6531	4	2.00
1001	7814	5	10.00
1001	2305	12	6.00
1002	7814	10	10.00
1002	7301	5	1.00
1003	4234	12	6.00
1003	2305	9	6.00
1004	6531	5	2.00
1004	2305	9	6.00

items		
item_id	item_desc	item_price
6531	Tape	2.00
7814	Paper	10.00
2305	Pens	6.00
7301	Erasers	1.00
4234	Pencils	6.00

```
create table normalization_activity.states (  
    state_id int primary key generated always as identity  
    , state_abbrev text not null unique  
    , state_name text not null unique  
);
```

```
create table normalization_activity.customers (  
    cust_id int primary key generated always as identity  
    , cust_name text not null  
    , state_id int  
        references normalization_activity.states (state_id)  
);
```

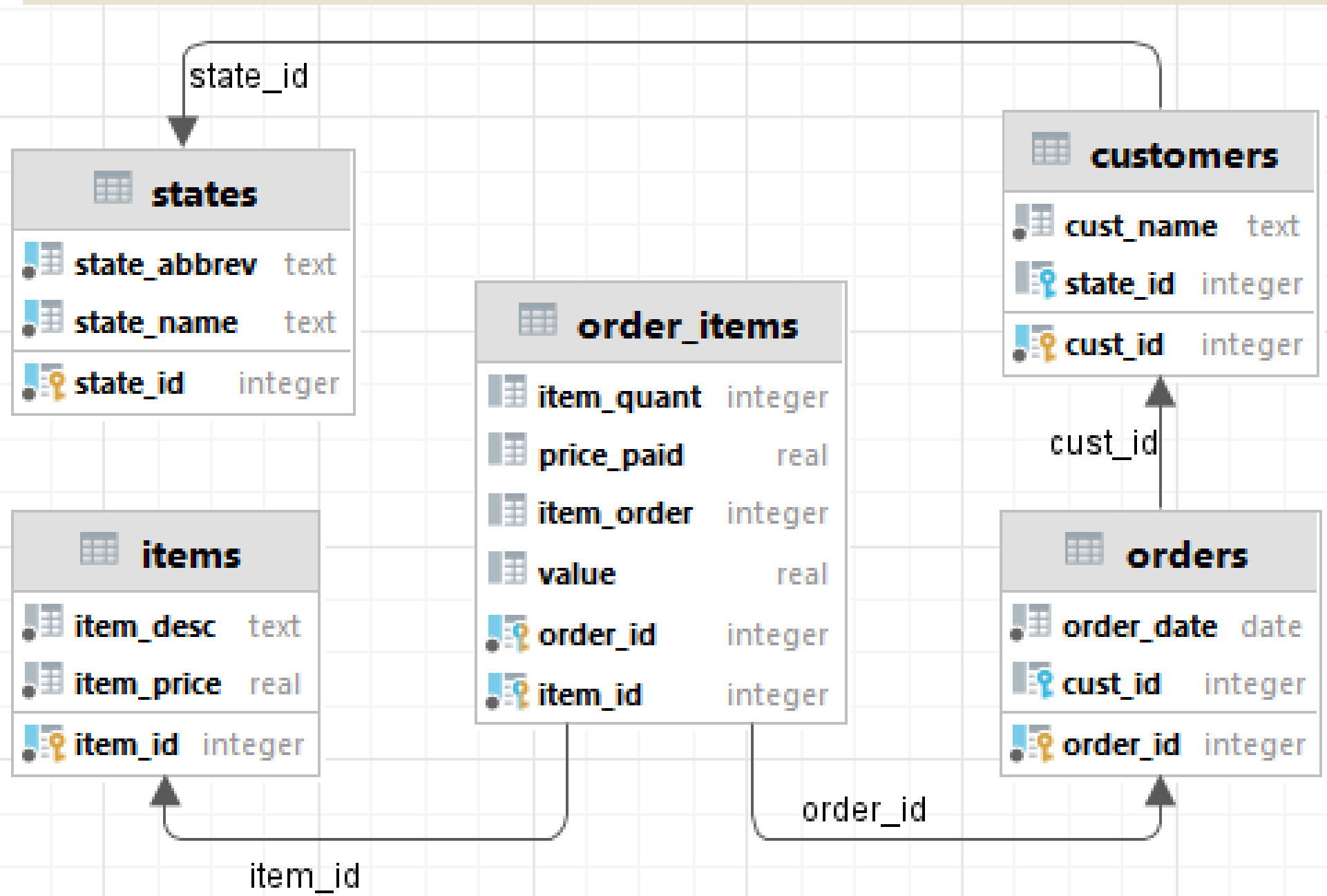
```
create table normalization_activity.orders (  
    order_id int primary key generated always as identity  
    , order_date date not null  
    , cust_id int  
        references normalization_activity.customers (cust_id)  
);
```

```
create table normalization_activity.items (  
    item_id int primary key generated always as identity  
    , item_description text not null  
    , item_price real default 0 not null  
);
```

```
create table normalization_activity.order_items (  
    order_id int  
        references normalization_activity.orders (order_id)  
    , item_id int  
        references normalization_activity.items (item_id)  
    , item_quant int  
    , price_paid real  
    , constraint order_items_pk  
        primary key (order_id, item_id)  
);
```

```
create table normalization_activity.order_items (  
    order_id int  
        references normalization_activity.orders (order_id)  
    , item_id int  
        references normalization_activity.items (item_id)  
    , item_quant int  
    , price_paid real  
    , item_order int  
    , constraint order_items_pk  
        primary key (order_id, item_id)  
);
```

```
create table normalization_activity.order_items (  
    order_id int  
        references normalization_activity.orders (order_id)  
    , item_id int  
        references normalization_activity.items (item_id)  
    , item_quant int  
    , price_paid real  
    , item_order int  
    , value real generated always  
        as (item_quant * price_paid) stored  
    , constraint order_items_pk  
        primary key (order_id, item_id)  
);
```



You are part owner of the **newest hot internet startup company.**

Your role in the company is database administrator and data analyst.

Your **7 figure salary** and **8 figure bonus** rely on you being able to keep the database up and running and then to query it for necessary information to make important business decisions.



I am the CEO asking you for information about our company and sales.



**YOU
GOT
THIS!**

I want to see a list of orders, order items (with items listed in the correct order) and customers. It should be ordered by date, customer id, and order id.

order_date	cust_name	state_abbrev	order_id	item_id	item_desc	item_quant	price_paid	value
2023-10-12	Larry	CA	1,002	7,814	Paper	10	10	100
2023-10-12	Larry	CA	1,002	7,301	Erasers	5	1	5
2023-10-13	Mo	OR	1,001	6,531	Tape	4	2	8
2023-10-13	Mo	OR	1,001	7,814	Paper	5	10	50
2023-10-13	Mo	OR	1,001	2,305	Pens	12	6	72
2023-10-17	Curly	WA	1,003	4,234	Pencils	12	6	72
2023-10-17	Curly	WA	1,003	2,305	Pens	9	2	18
2023-10-19	Mo	OR	1,004	6,531	Tape	5	2	10
2023-10-19	Mo	OR	1,004	2,305	Pens	9	6	54

I want to see a list of each order from all customers and the value of the order.

cust_name	order_id	order_date	order_value
Larry	1,002	2023-10-12	105
Mo	1,001	2023-10-13	130
Curly	1,003	2023-10-17	90
Mo	1,004	2023-10-19	64

I want to contact our most important customers and thank them for their business and ask how we can strengthen our relationship.

I want to see a list of each customer and how much each customer has ordered.

cust_name	cust_value
Mo	194
Larry	105
Curly	90

The business is thinking expanding our physical presence in areas where we have the most business.

I want to see the states where we have the greatest sales.

state_abbrev	state_value
OR	194
CA	105
WA	90

Some items are more important to be sure we have enough in inventory.

I want to see what items have the greatest value, how many are ordered, how many times they are ordered, and what is the value of that item.

item_id	item_desc	units_ordered	value	times_ordered
7,814	Paper	15	150	2
2,305	Pens	30	144	3
4,234	Pencils	12	72	1
6,531	Tape	9	18	2
7,301	Erasers	5	5	1